SCm Phonon-Coupled Inflationary Scale Factor with S26 Third-Order Ramanujan

Daniel T. Murphy

PAPER_1084: SCm Phonon-Coupled Inflationary Scale Factor with $S_{26}^{(3)}$ Ramanujan

Abstract

We derive the explicit phonon-coupled inflationary scale factor $a(t) = a_0 \exp(H_{\text{infl}} \cdot t)$ where the inflationary Hubble rate is:

$$H_{\text{infl}} = \sqrt{\frac{8\pi G}{3} \rho_{\text{SCm}}} \cdot S_{26}^{(3)}([\text{SSq}]) \cdot \Phi_{1.25\,\text{THz}}(\omega, \Gamma)$$

This calculator implements the 3rd-order Ramanujan summation $S_{26}^{(3)} = \sum_{n=1}^{26} e^{-[\text{SSq}] n / 26} \cdot n^3$ with Gaussian phonon resonance at 1.25 THz. Distinct from PAPER_1073 (which derives the full inflation cosmology) and PAPER_587 (which uses the Grind model), this paper provides the computational implementation with explicit $S_{26}^{(3)}$ and Gaussian Φ coupling.

§1 Inflationary Hubble Rate

$$H_{\text{infl}} = \sqrt{\frac{8\pi G}{3} \rho_{\text{SCm}}} \cdot S_{26}^{(3)} \cdot \Phi_{1.25\,\text{THz}}$$

§2 Third-Order Ramanujan Summation

$$S_{26}^{(3)} = \sum_{n=1}^{26} e^{-[\text{SSq}] \cdot n / 26} \cdot n^3$$

Numerical value: $S_{26}^{(3)} \approx 72{,}200$ for $[\text{SSq}] = 0.57$.

§3 Gaussian Phonon Resonance

$$\Phi_{1.25\,\text{THz}}(\omega, \Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}^{(3)}$$

At resonance ($\omega = \omega_{\text{SCm}}$): $\Phi = \Phi_0 \cdot S_{26}^{(3)} \approx 7.22 \times 10^{24}$.

§4 Scale Factor and E-Folds

$$a(t) = a_0 \exp(H_{\text{infl}} \cdot t)$$

$$N = H_{\text{infl}} \cdot t$$

For GUT-epoch values ($\rho_{\text{SCm}} \sim 10^{76}$ kg/m^3, $t \sim 10^{-32}$ s): $H_{\text{infl}} \sim 10^{12}$ s^{-1}, $N \sim 10^{-20}$ (requires GUT density for $N = 60$).

§5 Validation

Parameter	Value	Unit
H_{infl}	1.39×10^{12}	s^{-1}
$a(t = 10^{-32}\text{s})$	~ 1.0	(tiny e-folds at low density)
$S_{26}^{(3)}$	$72{,}212$	dimensionless
Φ (at resonance)	7.22×10^{24}	phonons/m^2/s

References

- PAPER_1073: SCm Phonon-Driven Inflation (full cosmology)
- PAPER_587: UQFFInflationaryEpochDetailsCalculator (Grind model)
- PAPER_1085: Phonon-Modulated Hubble Parameter

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